Leveraging HIV Surveillance Data to Improve Retention in Care

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NATIONAL RYAN WHITE CONFERENCE 2026 ON HIV CARE & TREATMENT

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Project Support

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Dr. Wiehe is a member of the U.S. Preventive Services Task Force (USPSTF). Materials provided in this presentation do not necessarily represent the views or recommendations of the USPSTF.

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Research Ethics

This study underwent full board review by Indiana University's (IU) Institutional Review Board and the Marion County Public Health Department's (MCPHD) Research Review Committee.

All team members underwent training in human participant research ethics, confidentiality, and requirements of the U.S. Health Insurance Portability and Accountability Act.

One team member is a public health lawyer and bioethics expert.

Our work is guided by community-engaged consultants.

Learning Objectives

Learning Objectives

After this activity, participants will be able to:

1. Describe how HIV surveillance or treatment data can be utilized to proactively identify people living with HIV who are at risk of falling out of care
2. Explain the development, testing, and refinement of the SAS-based predictive alert algorithm, including methods used to reduce unwarranted alerts and tailor outreach based on viral load status
3. Identify practical strategies for adapting or integrating the SAS code and eHARS-driven workflow into other HIV care systems, such as CareWARE or electronic medical records, to strengthen retention and relinkage efforts

Research Partners

Research Partners – IU School of Medicine

The IU School of Medicine's FairHealth Collective Team:

- Proposed the project and obtained grant funding
- Conducted community-engaged research to determine the need for, and methods of, the project
- Co-wrote and tested the Early Alert Retention in Care (RIC) Program code
- Coordinated project implementation
- Devised the project evaluation survey
- Will analyze and publish the evaluation
- Will disseminate program code and outcomes

INDIANA UNIVERSITY
SCHOOL OF MEDICINE
Children's Health Services Research

Research Partners – IU School of Medicine



Tammie McCaskill, MPH, CPH, ELS is an epidemiologist and scientific writer/editor with extensive experience in public health practice and research. She has been affiliated with the Ryan White Part A Indianapolis Transitional Grant Area's (TGA) Planning Council since 2012. She oversees the Early Alert RIC Project.

Sarah Wiehe, MD, MPH is a pediatrician and health services researcher. Her expertise is in community engagement, epidemiology, and linking clinical and non-clinical data to inform interventions to improve health among vulnerable populations. She is the Principal Investigator and enjoys engaging with people from communities to inform this work.



Research Partners – IU School of Medicine

Other team members include:



- Matthew Aalsma, PhD
- J. Dennis Fortenberry, MD, MS
- Ross Silverman, JD, MPH
- Marc Rosenman, MD
- Sami Gharbi, MS ECE
- Unai Miguel Andres, MS, MURP
- Ryan Healey, BS
- Elaine Cuevas, MPA
- The CTSI Research Jam Team

Research Partners – MCPHD

The MCPHD's Epidemiology Department:

- Provided data for program development and testing
- Tested the code on their server
- Will assist in program evaluation

The MCPHD's Ryan White HIV Services Program's Treatment Adherence and Engagement Liaison team:

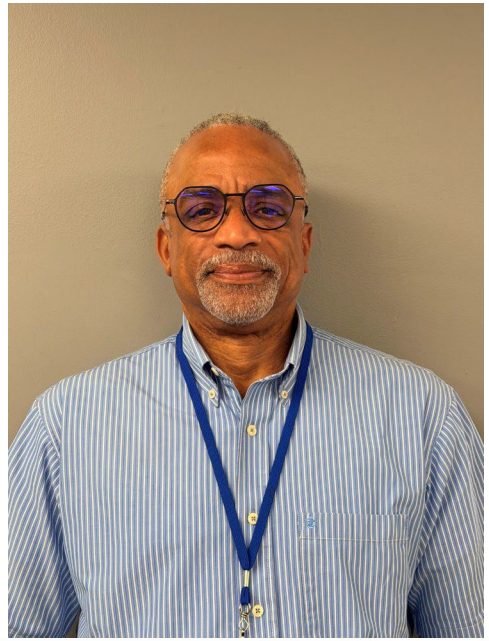
- Will utilize alerts to engage PLWH who are at risk of losing their RIC status
- Will collect data for project evaluation

MARION COUNTY PUBLIC HEALTH DEPARTMENT
Prevent. Promote. Protect.

Research Partners - MCPHD

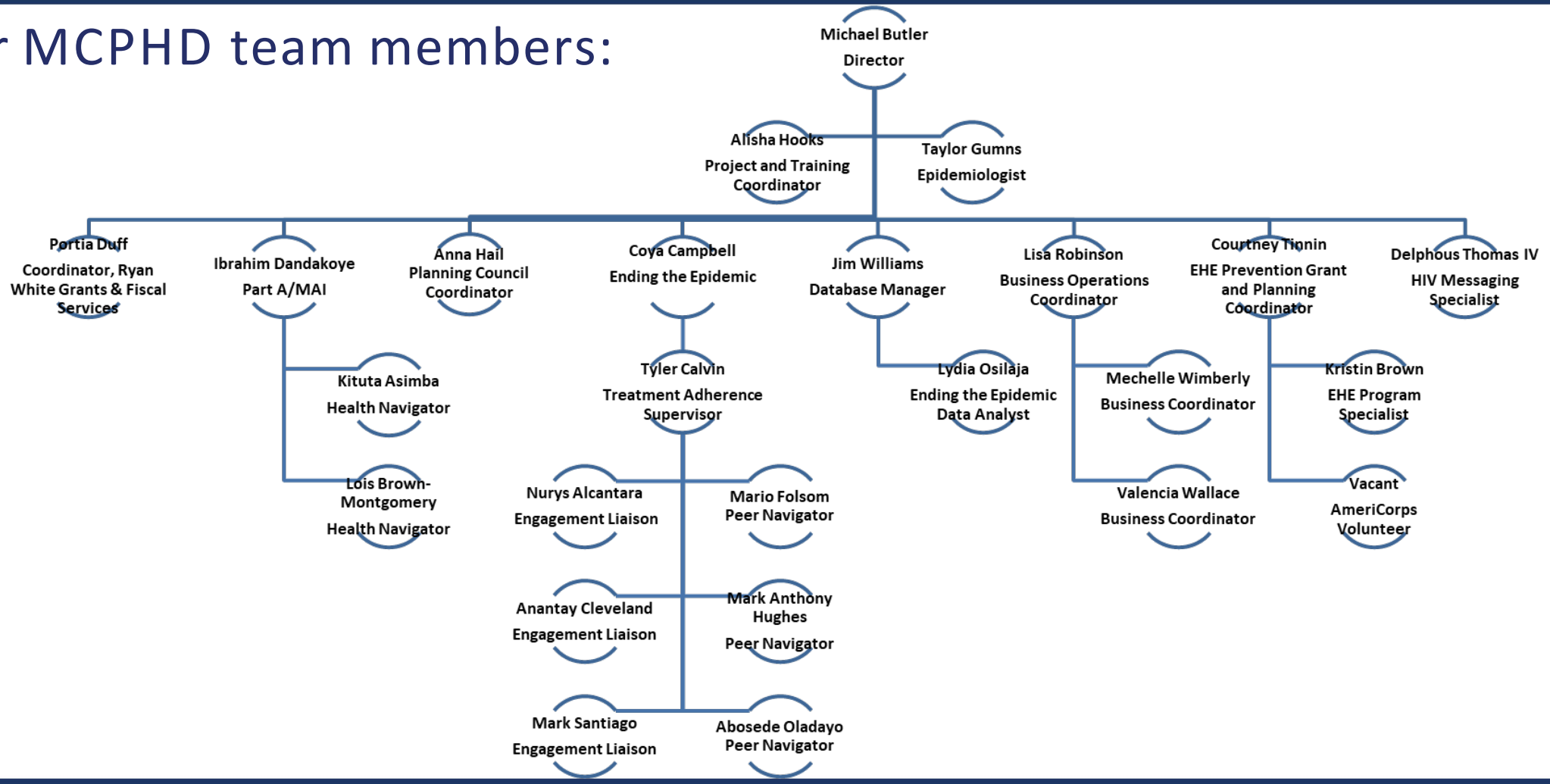
Taylor Gumns, MPH is an epidemiologist supporting both internal and external partners at MCPHD, as well as the Ryan White HIV Services Program and the Indianapolis Part A TGA.

Michael Butler, BS is Director of MCPHD's Ryan White HIV Services Program, managing the Indianapolis Part A Transitional Grant Area (TGA). He oversees all aspects of the program, including the Treatment Adherence and Engagement Liaison team.



Research Partners - MCPHD

Other MCPHD team members:



Research Partners – IU Biostatistics

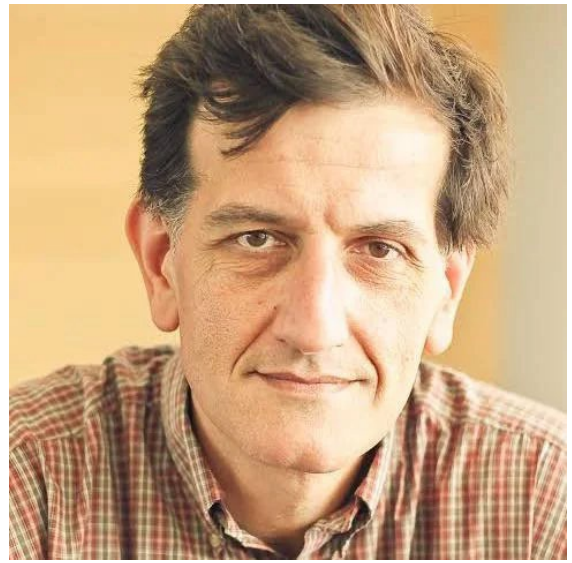
IU Richard M. Fairbanks School of Public Health's Department of Biostatistics and Health Data Science:

- Devised, co-wrote, and tested the Early Alert RIC Program
- Will assist in program evaluation and dissemination

BIOSTATISTICS AND HEALTH DATA SCIENCE
SCHOOL OF MEDICINE
RICHARD M. FAIRBANKS SCHOOL OF PUBLIC HEALTH

Research Partners – IU Biostatistics

Constantin Yiannoutsos, PhD is a professor of Biostatistics in the IU Fairbanks School of Public Health. He has spent over 25 years analyzing clinical and observational data obtained from people living with HIV (PLWH) in a variety of settings, both within the US and around the world. His primary research interest is developing methodologies related to loss to follow-up among PLWH, and he specializes in longitudinal data analysis.



Research Partners – State Health Department

The Indiana Department of Health's (IDOH) Division of HIV, STI, and Viral Hepatitis:

- Will train and contract with MCPHD to conduct engagement and linkage to care activities with individuals for whom an alert is generated
- Will run the Early Alert RIC Program against the state's HIV surveillance data
- Will conduct preliminary investigations to verify eligibility for case assignment
- Will assign eligible cases to MCPHD via the National Electronic Disease Surveillance System Base System (NBS)
- Will assist in program evaluation



Research Partners – Research Jam and Community Consultants

Our research was guided, from day one, by a large group of consultants. This consultant panel is made up of PLWH, clinicians who treat PLWH, and non-medical providers of services to PLWH, all in Marion County, Indiana (Indianapolis).[1]

Consultant engagement was facilitated by the Indiana Clinical and Translational Sciences Institute's Research Jam.

INDIANACTSI
Clinical and Translational Sciences Institute



Community Engagement

Community Engagement

Our community-engaged consultant panel made our research more relevant, impactful, and sustainable.[1]

- Our project was guided and informed by those who live with and/or work in HIV every day.
- Our community consultants laid the boundaries which should, or should not, be crossed to balance personal autonomy and respect with health outcomes and public health.
- Community engagement keeps our research person-centered.
- Community engagement empowers individuals living with HIV to collaborate meaningfully with our team.

Community Engagement

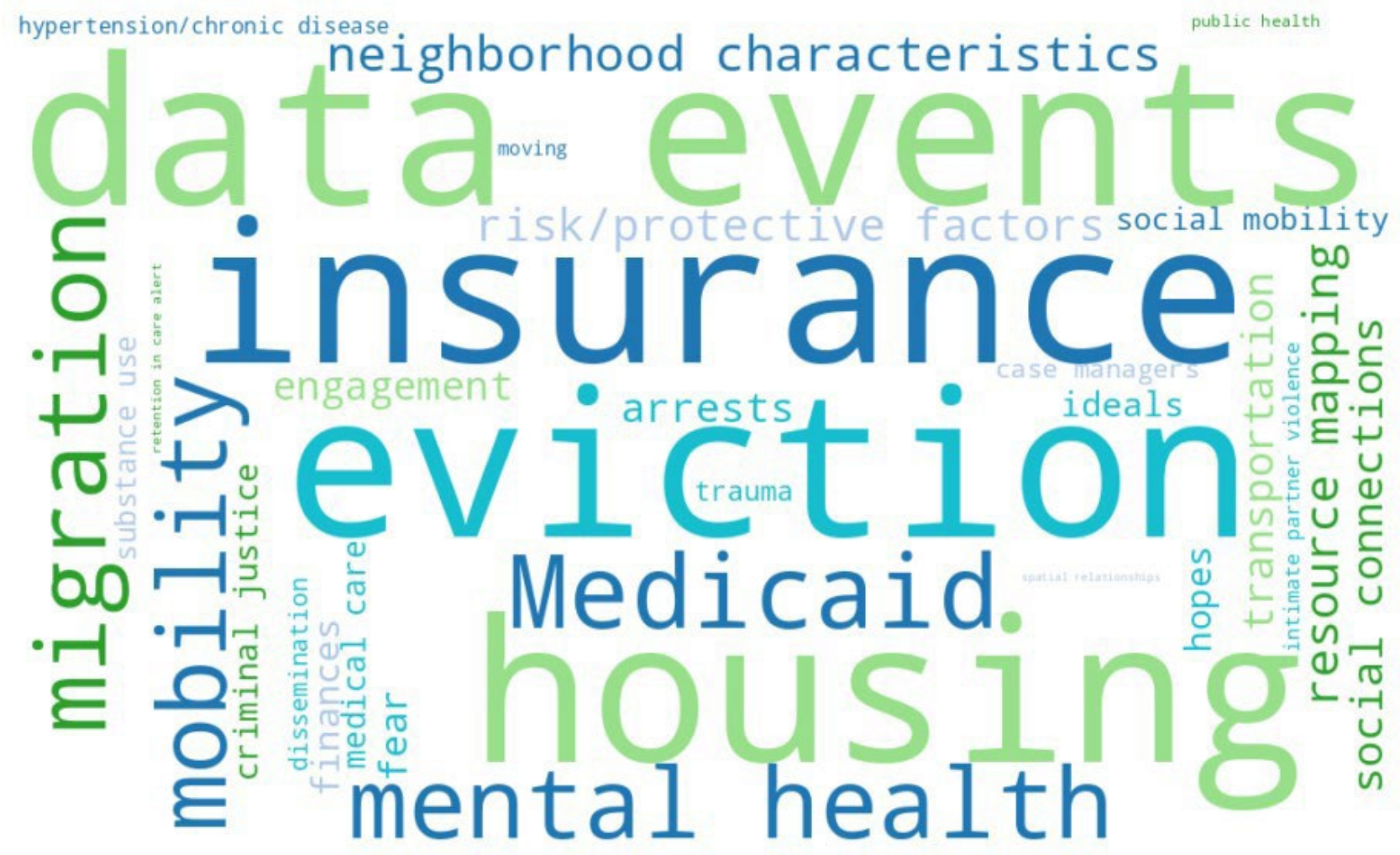
We hosted ten Research Jams (engagement workshops) for this project during 2020-2025.[1]

- Seventy-two (72) individuals participated on our consultant panel.
  - These individuals were largely representative of PLWH in Marion County by race/ethnicity.
  - The panel was over-represented by females (40% panel vs. <25% PLWH in Marion County).
  - Few young (18-24-year-old) people represented (note on self-report)

The Impetus for this Project

Why Are We Pursuing the Early Alert RIC Project?

During Jams, consultants discussed many topics that may affect the health outcomes of PLWH. We selected several projects recommended by our panel based on these Jam sessions.[1]



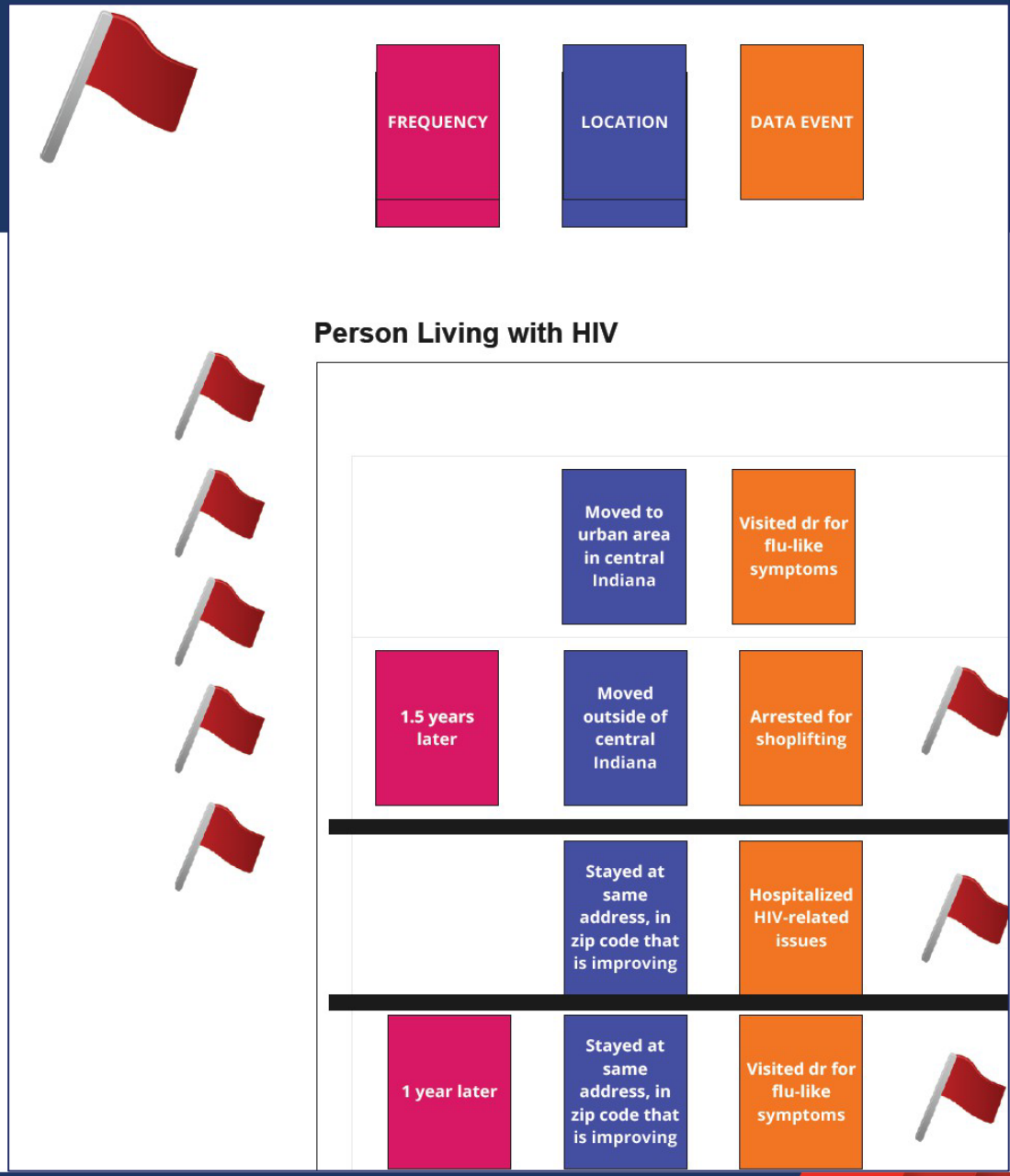
Why Are We Pursuing the Early Alert RIC Project?

We coordinated several activities to facilitate brainstorming among consultants. Three such activities led directly to the Early Alert RIC Project.[1]

- The Red Flag Game
- The Card Sorting Game
- The Reaching Out Decision Tree

The Red Flag Game

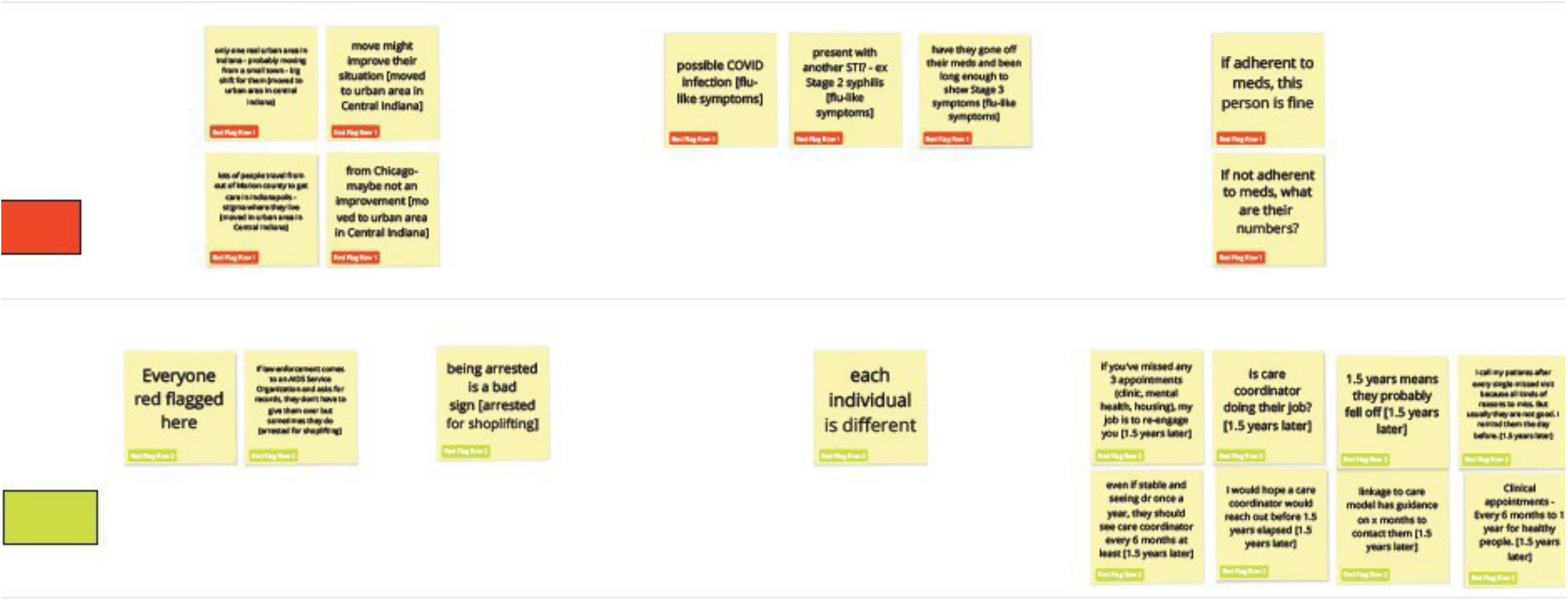
Consultants brainstormed events that might precede HIV-related health getting better or worse. They used red flags to explore when and for what reason(s) they would prioritize for reaching out to a patient/client following these events.



The Card Sorting Game

Results from the Red Flag Game were transferred to cards by frequency, location (e.g., rural area), and event (e.g., arrest). These cards were sorted according to perceived level of concern and under what circumstances public health and/or case managers should check in with a client.

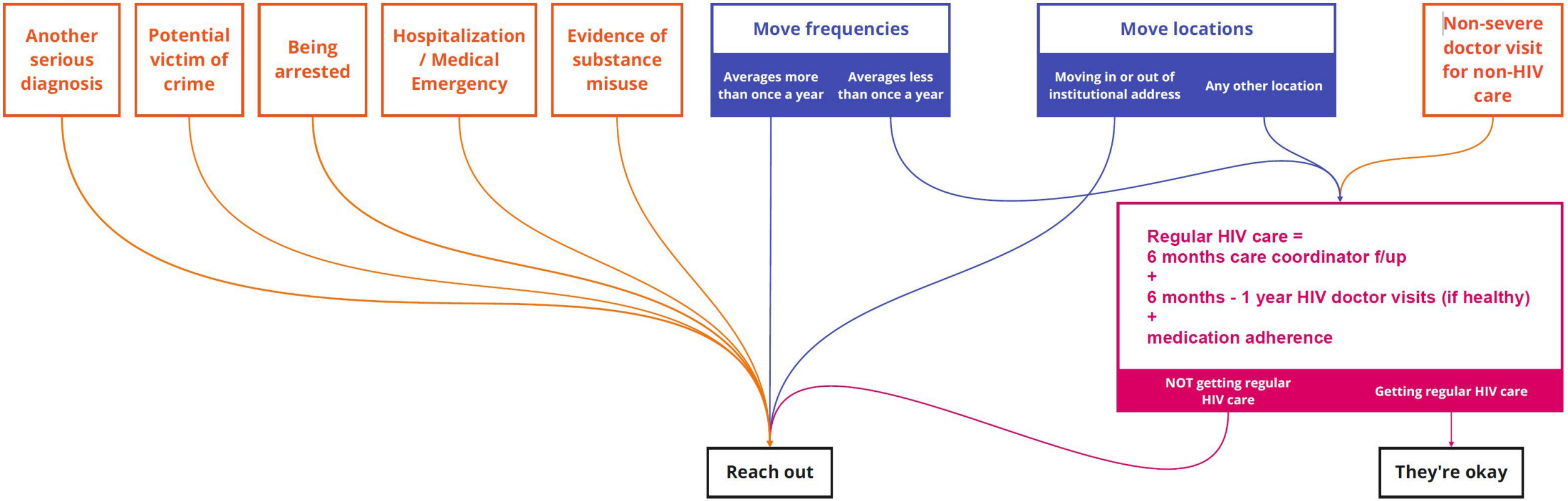
The Card Sorting Game



The Reaching Out Decision Tree

Results of these games were used to facilitate further discussion among our consultants and, eventually, their creation of a Reaching Out Decision Tree for use by our research team.

The Reaching Out Decision Tree



The Reaching Out Decision Tree

This product guided further discussion with our consultants, the MCPHD, the IDOH, the Ryan White Part A Planning Council, and Ryan White Part A medical and non-medical providers regarding our implementation of the Early Alert RIC Project.

Early Alert RIC Project Background, Vision, and Objectives

HIV Surveillance Data Description

Enhanced HIV/AIDS Reporting System (eHARS)

- A browser-based surveillance tool maintained by HHS's Centers for Disease Control and Prevention (CDC)[2]
- Contains demographics, HIV/AIDS diagnosis dates, HIV treatment and outcome measures (i.e., CD4 and viral load dates/results), and more for all PLWH in the U.S.
- Used by state and large local health departments for reporting, data management, analysis, and transfer of data to the CDC[2]

Local HIV Surveillance Data Utilization

The MCPHD utilizes eHARS data for the 10-county Ryan White Part A TGA

- Person and lab datasets curated by IDOH for MCPHD
- eHARS data used to create epidemiological reports
- eHARS data are also used to inform Ryan White Part A prioritization, allocation, and progress reports
- Early Alert RIC Program code will be run against eHARS to generate alerts that will be assigned to the Ryan White HIV Services Program's Treatment Adherence and Engagement Liaison team for relinkage services for PLWH at risk of falling out of care

Local Engagement Efforts Based on eHARS

Until now, engagement efforts have been informed by utilization of *retrospective* eHARS data for RE-linkage purposes

- RIC reports are processed 1-2 times annually to identify PLWH who no longer meet the definition of RIC (received ≥2 quantitative CD4 or VL tests performed ≥90 days apart during a 12-month period)[3]

Project Vision

An academic-public health partnership to make eHARS data more valuable for the engagement process by using proactive alerts generated ***before*** an individual loses RIC status.

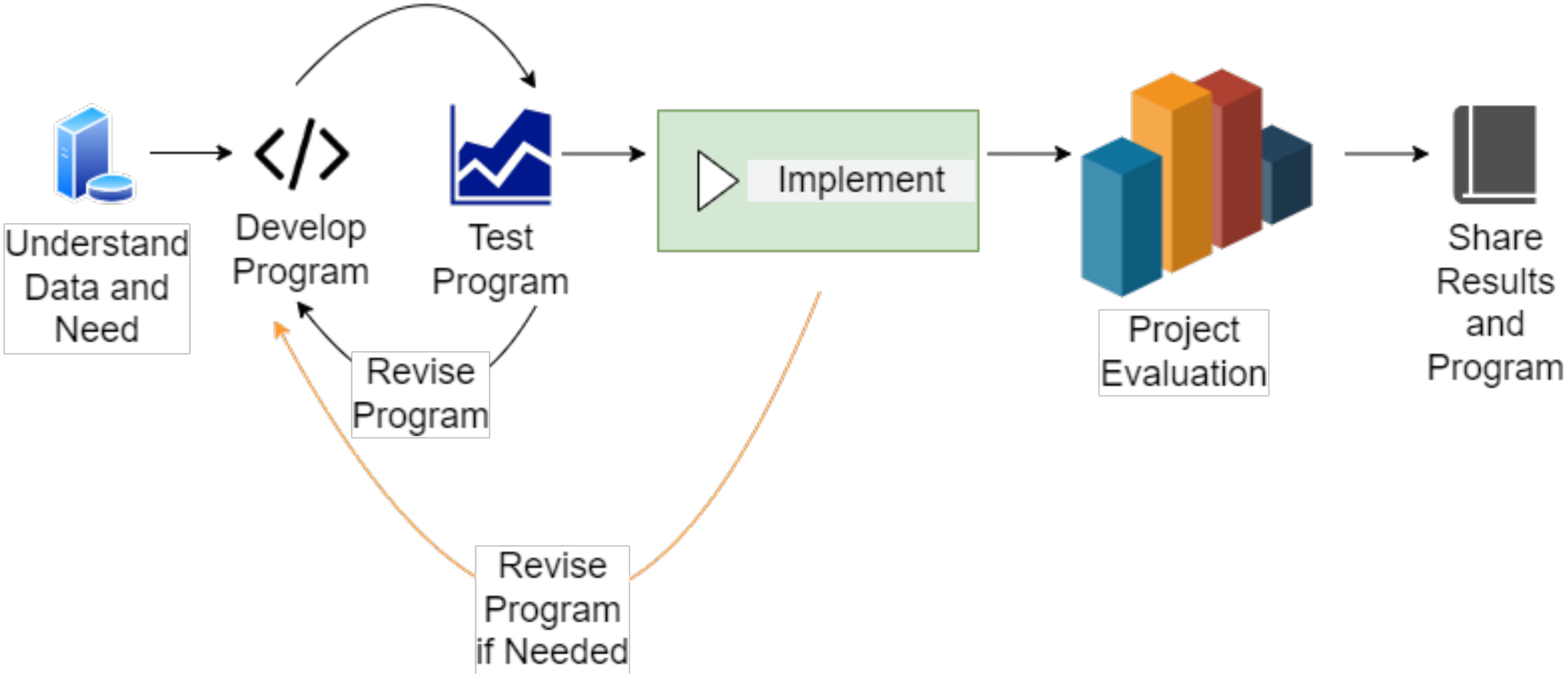
Project Objectives

IU, MCPHD, and IDOH will collaborate to:

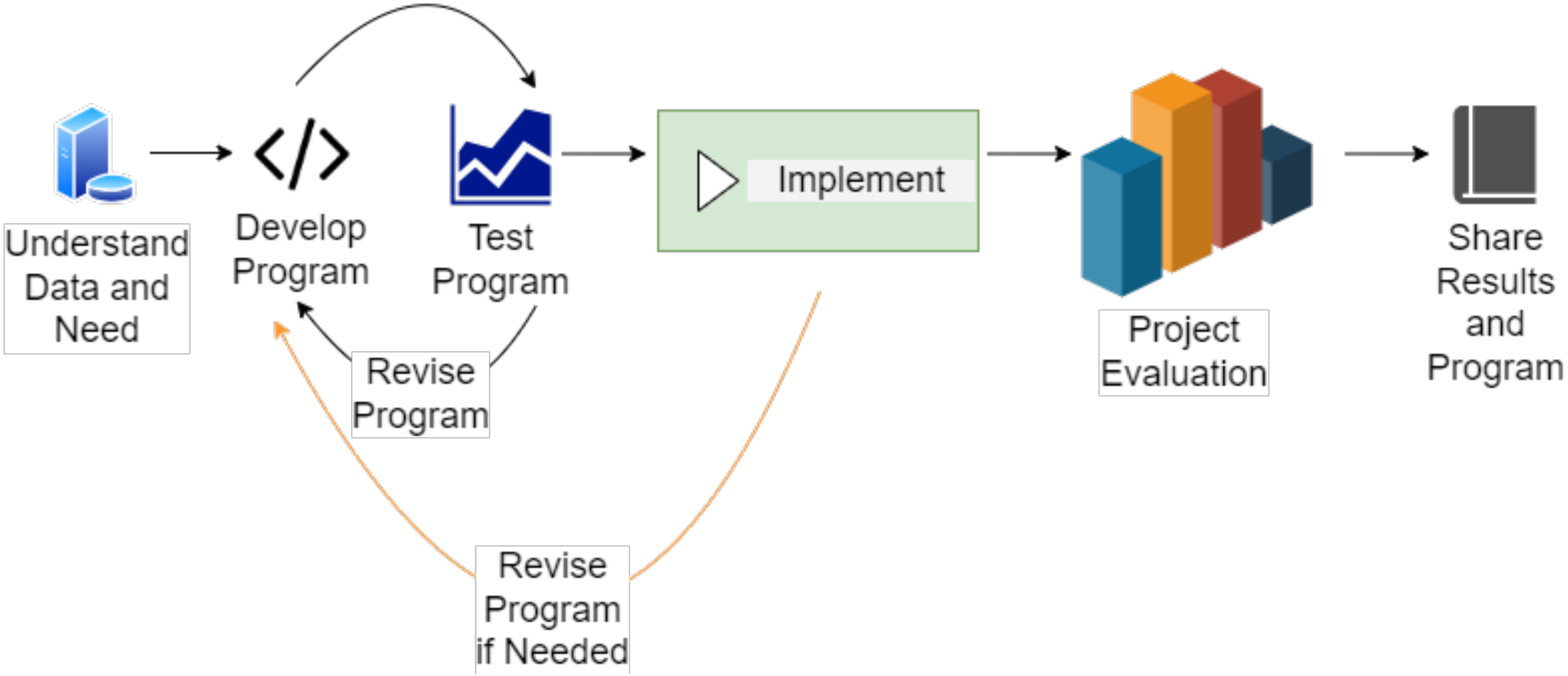
1. Create and test a program that generates proactive alerts to identify PLWH prior to them no longer having a RIC status
2. Conduct early engagement and linkage to care among PLWH to ensure they remain RIC
3. Evaluate and publish project outcomes
4. Obtain open-source licensing for the software and make it freely available for use by state and local health departments, and for adaptation for use against Ryan White's CAREWare and/or providers' electronic medical record systems

Software Development, Testing, and Revisions

Project Roadmap



Project Roadmap



Early Alert RIC Code v.1

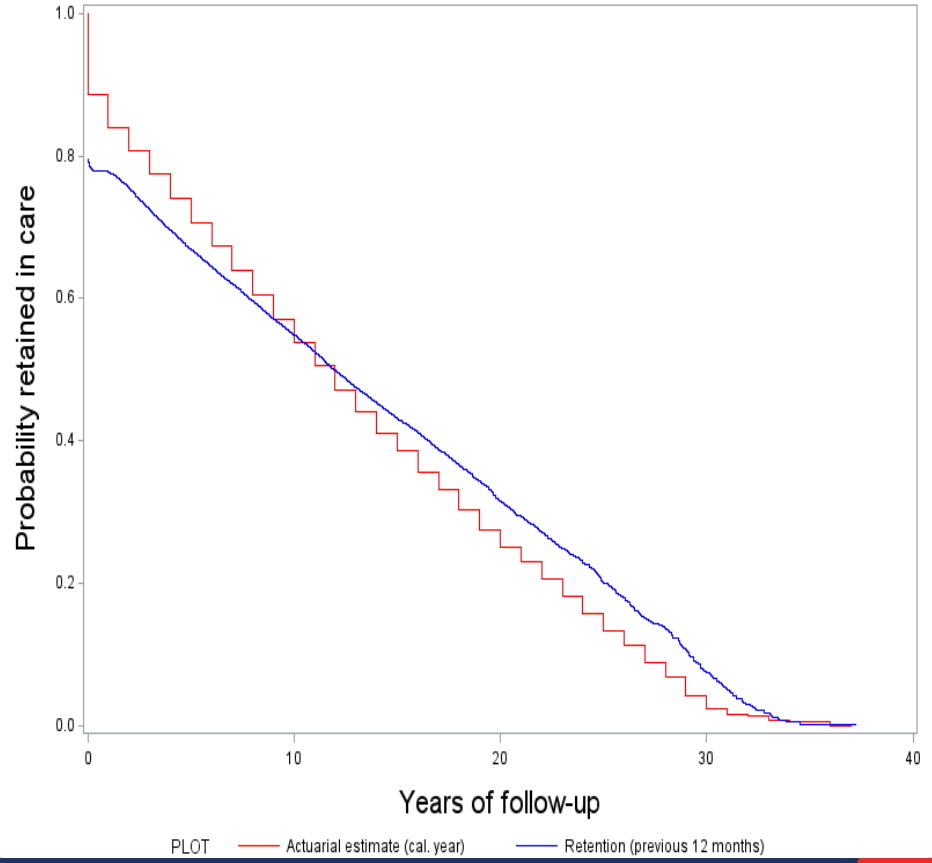
The Early Alert RIC Code v.1 looked backward 12 months to identify each individual's most recent quantitative CD4 or viral load test (qualifying lab) and then looked forward to confirm that each individual had received at least two qualifying labs ≥3 months apart.

Verifying the Algorithm

Early Alert RIC Code v.1

- Used to compare program results to that of the program currently in use
  - **Predicted outcomes aligned with those of the current method**, varying only due to the previous actuarial estimate (calendar year) vs. the monthly method used in our program

Comparison of HIV Care Attrition Using Estimates by Calendar Year vs. the Previous 12 Months in Marion County



Problems with Early Alert RIC Code v.1

Early Alert RIC Code v.1 was computationally difficult and resource intensive

- The code would run for many hours, tying up system resources.

This algorithm would not identify individuals who had no qualifying lab test prior to 12 months in the past, omitting individuals already failing to meet the definition of RIC.

Early Alert RIC Code v.2

Early Alert RIC Code v.2 utilized an algorithm that looked back to identify each individual's most recent qualifying lab and then looked forward from that date to calculate how long it had been since the qualifying lab, generating an alert for each individual 30 days prior to him or her losing RIC status (11 months from the most recent qualifying lab).

- This change greatly improved software efficiency with simplified and accelerated execution.
- It identified all PLWH who had lost, or were preparing to lose, RIC status.

Problems with Early Alert RIC Code v.2

Early Alert RIC Code v.2 identified everyone who had lost, or were at risk of losing, RIC status; however, it would have resulted in misused time because it created alerts:

- For deceased individuals
- For individuals no longer living in the TGA
- For individuals lost to follow up (out of care for many years) or not wanting to be contacted
- Regardless of the results of an individual's most recent viral load test

Early Alert RIC Code v.3 (Current Version)

Changes in the Early Alert RIC Code v.3 include:

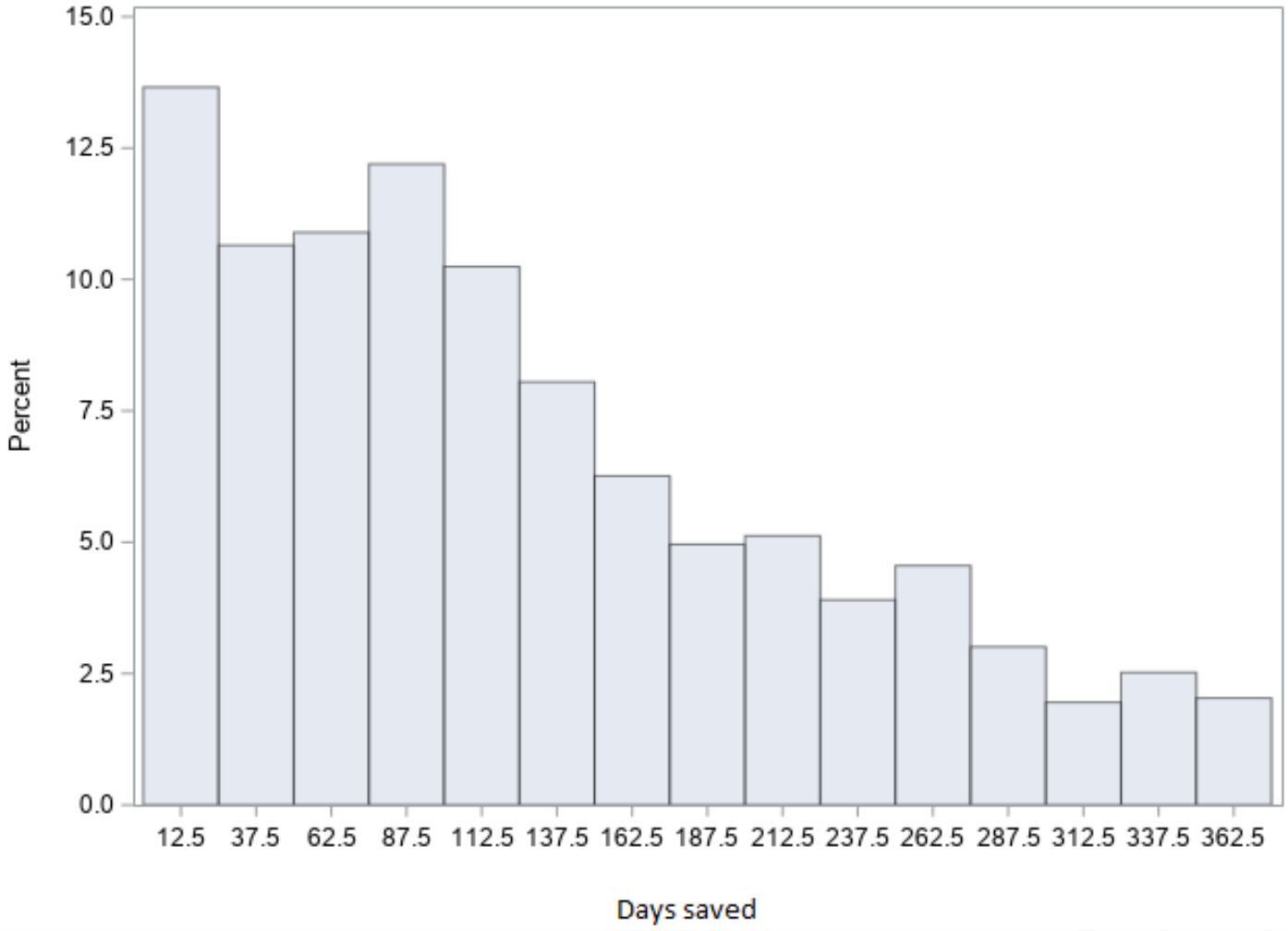
- Limits the look-back period to 24 months to shorten the alerts list
  - The look-back period can be increased or decreased using a macro within the code.
- Excludes individuals with an eHARS date of death
  - Can be linked to state death certificates if desired
- Excludes individuals known to have moved out of the jurisdiction
  - Must have been HIV/AIDS diagnosed and/or died in the jurisdiction *and* have received a lab in the new jurisdiction for the program to identify the new jurisdiction
- Creates alerts based on the most recent viral load result
  - Alerts 10 months after a detectable result (≥50 copies/mL) or 11 months after an undetectable result (<50 copies/mL)

Predicted Outcomes

Alerts generated for individuals at risk of losing RIC status will result in engagement before an individual falls out of care – in many cases, months earlier than using the current algorithm.

Predicted Outcomes

Based on historic (archived) eHARS data, the RIC Alert program will result in engagement efforts taking place about 3.5 months earlier than the method currently in use (Median = 106 days; IQR 52–190 days).



Program Implementation & Evaluation

Program Evaluation

eHARS data (i.e., post-alert qualifying lab dates) will be used to assess outcomes of engagement efforts among individuals for whom early alerts are generated.

Program Evaluation

MCPHD staff will also complete a survey for each case they work. These surveys are in REDCap and will include:

- Number of alerts that were unwarranted (e.g., incorrect, deceased)
- Number of alerts for which individuals are contacted
- Number of individuals unable to be contacted and/or connected to care
- Number of individuals connected to care prior to being non-RIC (from eHARS)
- Number of individuals connected to care within 30 days of falling out of care (from eHARS)
- Health outcomes following alert-based outreach (i.e., viral load) (from eHARS)
- Types of wraparound services referred to clients
- Time spent conducting linkage and engagement to those at risk of non-RIC

Program Implementation Steps Taken

- IU began the process of obtaining open-source licensing for the Early Alert RIC Program
- IU adapted the program for use by MCPHD and handed it over
- MCPHD tested the program on current eHARS data
- IU developed a project evaluation survey
- IU obtained IRB approval for project evaluation
- IU presented the project to IDOH and provided them with the program
- An agreement was negotiated for IDOH to train MCPHD to conduct engagement and linkage to care activities

Program Implementation Steps to be Taken

- IDOH will run the program against the state's HIV surveillance data bimonthly
- IDOH will conduct preliminary investigations against generated alerts to verify eligibility for case assignment (e.g., still alive, still living in Marion County)
- IDOH will assign eligible cases to MCPHD via NBS
- MCPHD will contact individuals assigned by IDOH to engage them into care and refer individuals to any needed wraparound services
- MCPHD will collect data for project evaluation
- IU, in collaboration with all participating organizations, will analyze and publish the evaluation
- IU will translate the code to R
- IU will disseminate program code in SAS and R

Next Steps

Next Steps

Routine medical care is critical to the health of PLWH and to ending the HIV epidemic.

Our academic-public health partnership shifts from retrospective to proactive (early) identification of PLWH at risk of falling out of care.

Our work will improve the health of PLWH nationwide when we share the code and program findings.

Questions?



Contact Us

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References

- https://ryanwhite.hrsa.gov/
- https://ryanwhite.hrsa.gov/about/recipient-resources
- Citations available upon request

Thank you!


Thank
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Gracias
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